

계량경제학 강의 13

자기상관 & 구조적 변화

1. Eviews 실습

- (1) Eviews로 인플레이션과 기대인플레이션의 그래프 그리기
- (2) 회귀분석: $\pi_t = \beta_1 + \beta_2 \pi_t^e + e_t$
- (3) 잔차항 그려보기
- (4) View → Correlogram 확인해보기
- (5) Equation 에서 View → residual diagnostics → Breusch-Godfrey serial correlation LM test → F 검정통계량과 p value 확인

(Auto / serial correlation) ✓

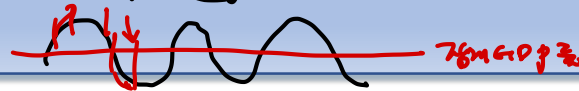
2. 자기상관모형

- 단항과 이항

- 공공성 (6744) ✓

- random

- GDP (777) 는. 불확



< Basic >

$$\underline{Y}_t = \beta X_t + \underline{e}_t, \quad e_t = \rho e_{t-1} + v_t, \quad v_t \sim iid(0, \sigma^2)$$

$e_t \sim iid N(0, \sigma^2)$ X

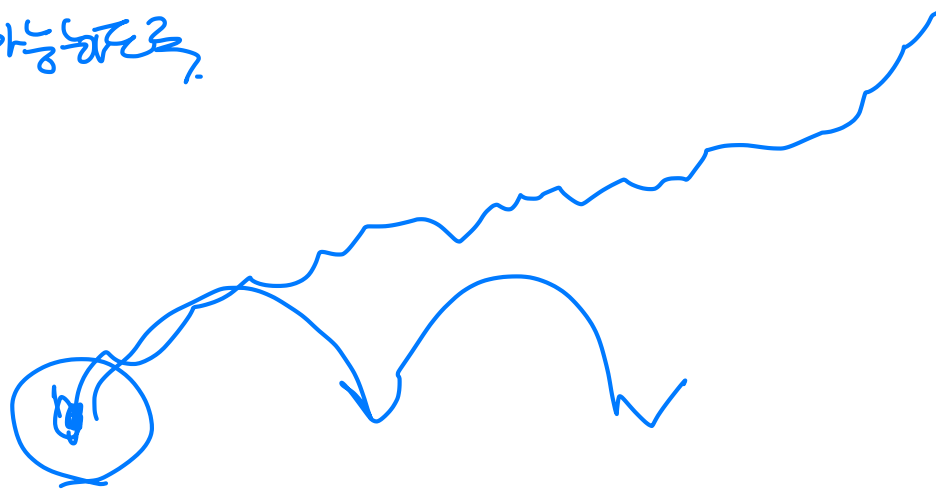
Unpredictable

$$E(e_t \cdot e_s) = 0, \quad \forall t \neq s$$

• GLS의 아이디어:

이분산 모형과 마찬가지로 고전적 가정이 성립하도록 모형을 변형시켜 OLS를 적용하자.

1b. (*) $\Rightarrow$ OLS가 가능해짐



2. 자기상관모형

단,

$$|\rho| < 1$$

영수자 충족 ✓

일시적 충족 ✓

Known

- 분석을 간단히 하기 위해 ρ 가 알려져 있는 경우를 생각해보자.

$$Y_t = \beta X_t + e_t, \quad e_t = \rho e_{t-1} + v_t, \quad v_t \sim iid(0, \sigma^2)$$

$$-\rho Y_{t-1} = \rho \beta X_{t-1} + \rho e_{t-1}$$

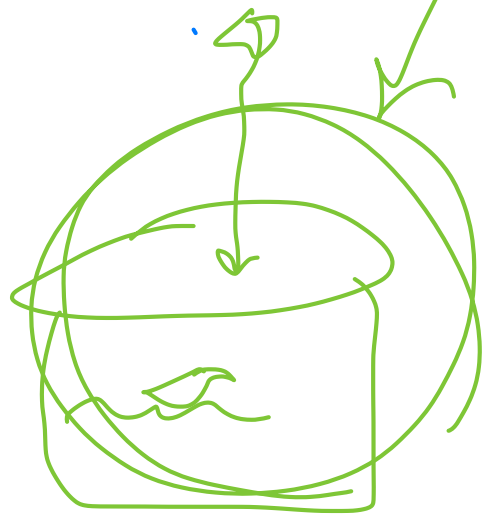
lagged

$$Y_t - \rho Y_{t-1} = \beta (X_t - \rho X_{t-1}) + (e_t - \rho e_{t-1})$$

$$\rightarrow Y_t^* = \beta X_t^* + e_t^*$$

$$\Rightarrow \hat{\rho} \text{ 찾는?}$$

How to Estimate?



3. Cochrane-Orcutt iterative Procedure

- ρ 가 알려져 있지 않은 경우

(1) OLS regression : $Y_t = \beta X_t + e_t \rightarrow \text{get } \hat{\beta} \text{ \& } \hat{e}_t$

(2) OLS regression : $\hat{e}_t = \rho \hat{e}_{t-1} + v_t \rightarrow \text{get } \hat{\rho}$ ✓

(3) GLS regression with $\hat{\rho}$:

$$Y_t^* = \beta X_t^* + v_t \quad (Y_t^* = Y_t - \hat{\rho} Y_{t-1}, X_t^* = X_t - \hat{\rho} X_{t-1}) \rightarrow \text{get } \hat{\beta} \text{ \& } \hat{e}_t$$

(4) Go back to (2)

수렴할 때까지 반복

Note

$e_t = \rho_1 e_{t-1} + \rho_2 e_{t-2} + v_t$
 First-order Autocorrelation
 $\Leftrightarrow$ Autoregressive of order 1
 $\Rightarrow$ AR(1)

$e_t = \rho_1 e_{t-1} + \rho_2 e_{t-2} + v_t$
 $\Rightarrow$ AR(2)

✓ $\hat{\rho}$ off autocorrelation

4. 구조적 변화가 있는 경우

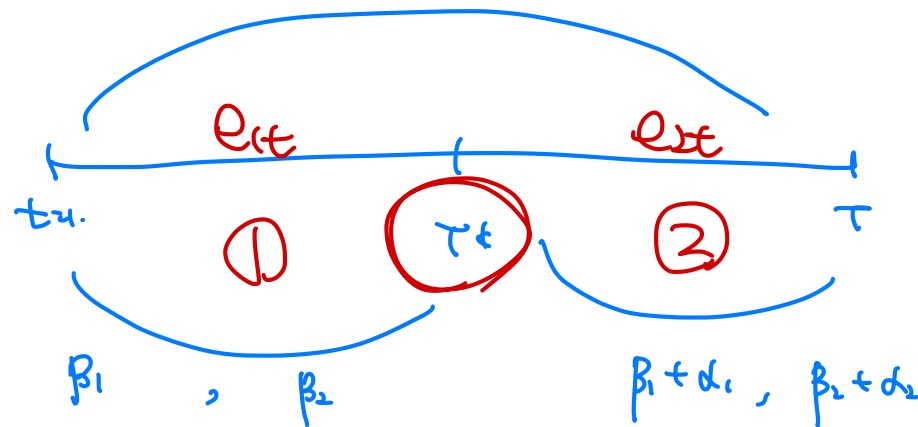
- 경제에 구조적 변화(structural change)가 있는지 test 하는 경우

: Dummy variable 을 이용하여 나타낼 수 있다

$$Y_t = (\beta_1 + \alpha_1 D_t) + (\beta_2 + \alpha_2 D_t) X_{2t} + e_t, \quad e_t \sim (0, \sigma^2)$$
$$= \gamma_1 + \gamma_2 X_{2t} + e_t$$

Where $D_t = \begin{cases} 0, & t \leq T^* \\ 1, & t > T^* \end{cases}$

dormant



break-point test

4. 구조적 변화가 있는 경우

- 만일 구조적 변화가 없다면, $\beta_1 = \gamma_1$, $\beta_2 = \gamma_2$ 일 것이다.

- Chow test : $H_0: \beta_1 = \gamma_1, \beta_2 = \gamma_2$

회귀분석을 3번 실시

①

4. 구조적 변화가 있는 경우

귀무가설이 맞다면

$$\sum_{t=1}^T \widehat{e_t^{*2}} \approx \underbrace{\sum_{t=1}^{T^*} \widehat{e_{1t}^2}}_{\text{①}} + \underbrace{\sum_{t=T^*+1}^T \widehat{e_{2t}^2}}_{\text{②}}$$

$$f\text{-value} = \frac{(\sum \widehat{e_t^{*2}} - \sum \widehat{e_t^2})/2}{(T-4)} \sim F(2, T-4) \quad \checkmark$$

✓ T^k : known ✓

$$\textcircled{1} E(e_t) = ? \quad \textcircled{2} E(e_t \cdot e_{t-s}) = ? \quad \textcircled{3} E(e_t^2) = ?$$

↓

$$e_t = \rho \underline{e_{t-1}} + v_t$$

$$= \rho (\rho \underline{e_{t-2}} + v_{t-1}) + v_t$$

$$= \rho^2 \underline{e_{t-2}} + \rho \cdot v_{t-1} + v_t$$

$$= \rho^2 (\rho \cdot e_{t-3} + v_{t-2}) + \rho \cdot v_{t-1} + v_t$$

$$= \rho^3 \underline{e_{t-3}} + \rho^2 v_{t-2} + \rho \cdot v_{t-1} + v_t$$

$$= \rho^j \underline{e_{t-j}} + \dots + \dots + v_t + \rho \cdot v_{t-1} + \dots + \dots$$

$$j \rightarrow \infty$$

$$= V_t + \rho V_{t-1} + \rho^2 V_{t-2} + \dots + \rho^{\infty} V_{t-\infty} + \dots$$

$$|\rho| < 1$$

$$\rho^{\infty} \rightarrow 0$$

$$= \underline{V_t} + \underline{\rho V_{t-1}} + \underline{\rho^2 V_{t-2}} + \dots$$

$$V_t \sim \text{iid}(0, \sigma^2)$$

$$E(e_t) = 0$$

$$\textcircled{2} E(e_t^2) = ?$$

$$= \text{Var}(e_t)$$

(4, 4)

$$= \text{Var}(V_t) + \rho^2 \text{Var}(V_{t-1}) + \rho^4 \text{Var}(V_{t-2}) + \dots$$

$$e_t = \rho \cdot e_{t-1} + V_t$$

$$= \sigma^2 + \rho^2 \cdot \sigma^2 + \rho^4 \cdot \sigma^2 + \dots$$

$$= \sigma^2 (1 + \rho^2 + \rho^4 + \rho^6 + \dots)$$

$$= \frac{\sigma^2}{1 - \rho^2}$$

$$|\rho| < 1$$

$$\textcircled{2} \quad E(e_t \cdot e_{t-s}) = ?$$

$$\Rightarrow \text{Cov}(e_t, e_{t-s})$$

$$= E \left[(v_t + \rho \cdot v_{t-1} + \rho^2 v_{t-2} + \dots + \rho^s v_{t-s} + \rho^{s+1} v_{t-s-1} + \dots) (v_{t-s} + \rho \cdot v_{t-s-1} + \rho^2 v_{t-s-2} + \dots) \right]$$

$$= \rho^S \sigma^2 + \rho^{S+2} \sigma^2 + \rho^{S+4} \sigma^2 + \dots$$

$$= \sigma^2 \left(\rho^S + \rho^{S+2} + \rho^{S+4} + \dots \right)$$

$$= \sigma^2 \cdot \underbrace{\rho^S}_{\sim} \frac{1}{1 - \rho^2}$$

$$S \uparrow \text{대}, \quad \rho^S \rightarrow 0 \quad \text{if } |\rho| < 1$$

$$S \rightarrow \infty, \quad \rho^S = 0 \quad \text{if } |\rho| < 1$$

- 시계열의 수렴을 보장하기 위하여